JavaScript Built-in Functions

Complete Reference Guide

Strings . Arrays . Math . Numbers . Objects . JSON . Dates . Globals

83	8	100%
Functions	Categories	Free to Use

This guide covers every major built-in function available natively in JavaScript - no imports, no packages required. Each entry shows **what** the function does, **where** you use it in real projects, and a concise **code example**. Use it as a quick-reference beside your editor.

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String Functions (17)

`.length`

What: Returns number of characters in a string

Use: *Validate input length*

```
"hello".length        // 5
```

`.toUpperCase()`

What: Converts all characters to uppercase

Use: *Normalize text for comparison*

```
"hello".toUpperCase()  // "HELLO"
```

`.toLowerCase()`

What: Converts all characters to lowercase

Use: *Case-insensitive matching*

```
"HELLO".toLowerCase()  // "hello"
```

`.trim()`

What: Removes whitespace from both ends

Use: *Clean user form input*

```
"  hi  ".trim()        // "hi"
```

`.includes()`

What: Returns true if substring is found

Use: *Check if string contains value*

```
"hello".includes("ell") // true
```

`.startsWith()`

What: Returns true if string begins with value

Use: *URL or prefix validation*

```
"hello".startsWith("he")// true
```

`.endsWith()`

What: Returns true if string ends with value

Use: *File extension checks*

```
"photo.png".endsWith(".png") // true
```

`.slice()`

What: Extracts part of string by start/end index

Use: *Get substrings*

```
"hello".slice(1,3)      // "el"
```

`.split()`

What: Splits string into array by separator

Use: *Parse CSV, URLs, sentences*

```
"a,b,c".split(",")      // ["a","b","c"]
```

`.replace()`

What: Replaces first matching pattern

Use: *Simple text substitution*

```
"hi hi".replace("hi","bye") // "bye hi"
```

`.replaceAll()`

What: Replaces all matching patterns

Use: *Global text substitution*

```
"hi hi".replaceAll("hi","bye") // "bye bye"
```

`.indexOf()`

What: Returns index of first occurrence (-1 if none)

Use: *Find position of substring*

```
"hello".indexOf("l")    // 2
```

`.repeat()`

What: Returns string repeated n times

Use: *Build separator or padding lines*

```
"ab".repeat(3)          // "ababab"
```

`.padStart()`

What: Pads beginning to reach target length

Use: *Format numbers like 01, 02, 03*

```
"5".padStart(3,"0")     // "005"
```

`.padEnd()`

What: Pads end to reach target length

Use: *Fixed-width text columns*

```
"hi".padEnd(5,".")      // "hi..."
```

`.charAt()`

What: Returns character at given index

Use: *Read a specific character*

```
"hello".charAt(1)       // "e"
```

`.charCodeAt()`

What: Returns Unicode code at given index

Use: *Encoding or encryption logic*

```
"A".charCodeAt(0)       // 65
```

Array Functions (21)

`.push()`

What: Adds one or more items to end of array

Use: *Append items to a list*

```
let a=[1,2]; a.push(3); // [1,2,3]
```

`.pop()`

What: Removes and returns last item

Use: *Stack (LIFO) data structure*

```
[1,2,3].pop()           // 3
```

`.shift()`

What: Removes and returns first item

Use: *Queue (FIFO) operations*

```
[1,2,3].shift()         // 1
```

`.unshift()`

What: Adds items to beginning of array

Use: *Prepend to a list*

```
let a=[2,3]; a.unshift(1); // [1,2,3]
```

`.map()`

What: Creates new array by transforming each item

Use: *Transform data lists*

```
[1,2,3].map(x=>x*2)     // [2,4,6]
```

`.filter()`

What: Creates new array with items passing a test

Use: *Filter search results*

```
[1,2,3,4].filter(x=>x>2)// [3,4]
```

`.reduce()`

What: Reduces array to a single accumulated value

Use: *Sum totals, combine values*

```
[1,2,3].reduce((a,b)=>a+b,0) // 6
```

`.find()`

What: Returns first item that passes test

Use: *Find a user or record by ID*

```
[1,2,3].find(x=>x>1)    // 2
```

`.findIndex()`

What: Returns index of first matching item

Use: *Locate item position in list*

```
[1,2,3].findIndex(x=>x>1)// 1
```

`.includes()`

What: Checks if value exists in array

Use: *Membership / permission checks*

```
[1,2,3].includes(2)     // true
```

`.join()`

What: Joins all elements into a string

Use: *Build CSV or display strings*

```
[1,2,3].join("-")        // "1-2-3"
```

`.slice()`

What: Returns shallow copy of portion of array

Use: *Pagination, copy segments*

```
[1,2,3,4].slice(1,3)    // [2,3]
```

`.splice()`

What: Adds or removes items in place (mutates array)

Use: *Edit list in-place*

```
let a=[1,2,3]; a.splice(1,1); // [1,3]
```

`.sort()`

What: Sorts array in place

Use: *Sort names, numbers, dates*

```
[3,1,2].sort((a,b)=>a-b)// [1,2,3]
```

`.reverse()`

What: Reverses array in place

Use: *Show newest items first*

```
[1,2,3].reverse()       // [3,2,1]
```

`.flat()`

What: Flattens nested arrays by depth

Use: *Normalize nested data*

```
[[1,2],[3]].flat()      // [1,2,3]
```

`.forEach()`

What: Calls a function for each element

Use: *Logging, DOM updates (side effects)*

```
[1,2].forEach(x=>console.log(x))
```

`.every()`

What: Returns true if ALL items pass test

Use: *Form validation checks*

```
[2,4,6].every(x=>x%2===0) // true
```

`.some()`

What: Returns true if ANY item passes test

Use: *Check permissions list*

```
[1,2,3].some(x=>x>2)    // true
```

`Array.from()`

What: Creates array from any iterable

Use: *Convert NodeList or Set to array*

```
Array.from('abc')       // ["a","b","c"]
```

`Array.isArray()`

What: Returns true if value is an array

Use: *Type checking before operations*

```
Array.isArray([1,2])    // true
```

Math Functions (12)

`Math.round()`

What: Rounds to nearest integer

Use: *Display prices, scores*

```
Math.round(4.6)         // 5
```

`Math.floor()`

What: Rounds down to nearest integer

Use: *Pagination page numbers*

```
Math.floor(4.9)         // 4
```

`Math.ceil()`

What: Rounds up to nearest integer

Use: *Calculate rows needed in a grid*

```
Math.ceil(4.1)          // 5
```

`Math.abs()`

What: Returns absolute (positive) value

Use: *Distance, difference calc*

```
Math.abs(-5)            // 5
```

`Math.max()`

What: Returns largest value from list

Use: *Find highest score or price*

```
Math.max(1,5,3)         // 5
```

`Math.min()`

What: Returns smallest value from list

Use: *Find lowest price*

```
Math.min(1,5,3)         // 1
```

`Math.pow()`

What: Raises base to the given exponent

Use: *Compound interest, area formulas*

```
Math.pow(2,8)           // 256
```

`Math.sqrt()`

What: Returns the square root of a number

Use: *Distance formula, geometry*

```
Math.sqrt(16)          // 4
```

`Math.random()`

What: Returns random float between 0 and 1

Use: *Games, shuffle, random picks*

```
Math.floor(Math.random()*10) // 0-9
```

`Math.trunc()`

What: Removes decimal part without rounding

Use: *Integer division*

```
Math.trunc(4.9)        // 4
```

`Math.sign()`

What: Returns 1, -1, or 0 based on sign of number

Use: *Determine direction or polarity*

```
Math.sign(-5)          // -1
```

`Math.PI`

What: Value of pi: 3.14159...

Use: *Circle area and circumference*

```
Math.PI                // 3.14159...
```

Number Functions (8)

`Number()`

What: Converts a value to number type

Use: *Parse user input strings*

```
Number("42")           // 42
```

`.toFixed(n)`

What: Rounds to n decimal places, returns string

Use: *Display currency values*

```
(3.14159).toFixed(2)    // "3.14"
```

`.toString(base)`

What: Converts number to string (optional base)

Use: *Binary or hex representation*

```
(255).toString(16)      // "ff"
```

`Number.isInteger()`

What: Returns true if value is an integer

Use: *Input validation*

```
Number.isInteger(4)     // true
```

`Number.isNaN()`

What: Returns true if value is NaN

Use: *Catch bad parse results*

```
Number.isNaN(NaN)       // true
```

`Number.isFinite()`

What: Returns true if value is finite

Use: *Prevent Infinity in calculations*

```
Number.isFinite(Infinity)// false
```

`parseInt()`

What: Parses string and returns integer

Use: *Read integer from text input*

```
parseInt("42px")        // 42
```

`parseFloat()`

What: Parses string and returns float

Use: *Read decimal from text input*

```
parseFloat("3.14abc")   // 3.14
```

Object Functions (7)

`Object.keys()`

What: Returns array of object's own property names

Use: *Iterate over object properties*

```
Object.keys({a:1,b:2}) // ["a","b"]
```

`Object.values()`

What: Returns array of object's own property values

Use: *Sum or display all values*

```
Object.values({a:1,b:2})// [1,2]
```

`Object.entries()`

What: Returns array of [key, value] pairs

Use: *Convert object to iterable list*

```
Object.entries({a:1})  // [["a",1]]
```

`Object.assign()`

What: Copies properties from sources to target

Use: *Merge config objects*

```
Object.assign({},{a:1},{b:2}) // {a:1,b:2}
```

`Object.freeze()`

What: Prevents modification of object

Use: *Immutable constants / config*

```
const o=Object.freeze({a:1})
```

`Object.hasOwn()`

What: Returns true if object has own property

Use: *Safe property existence check*

```
Object.hasOwn({a:1},"a")// true
```

`typeof`

What: Returns type of value as string

Use: *Type guards before operations*

```
typeof "hi"             // "string"
```

JSON Functions (2)

`JSON.stringify()`

What: Converts JS value to JSON string

Use: *Send data to API / localStorage*

```
JSON.stringify({a:1})   // '{"a":1}'
```

`JSON.parse()`

What: Parses JSON string into JS value

Use: *Read API responses / stored data*

```
JSON.parse('{"a":1}')  // {a: 1}
```

Date Functions (6)

`new Date()`

What: Creates a new Date object

Use: *Get current date and time*

```
new Date()              // Thu Apr 23 2026
```

`.getFullYear()`

What: Returns 4-digit year

Use: *Display or compare years*

```
new Date().getFullYear()// 2026
```

`.getMonth()`

What: Returns month index (0-11)

Use: *Month-based logic (+1 for display)*

```
new Date().getMonth()   // 3 (April)
```

`.getDate()`

What: Returns day of month (1-31)

Use: *Show current day number*

```
new Date().getDate()    // 23
```

`.toLocaleDateString()`

What: Returns locale-formatted date string

Use: *Display readable date to users*

```
new Date().toLocaleDateString() // "4/23/2026"
```

`Date.now()`

What: Returns current timestamp in ms

Use: *Measure elapsed time, unique IDs*

```
Date.now()              // 1745366400000
```

Global Functions (10)

`console.log()`

What: Prints output to browser developer console

Use: *Debug during development*

```
console.log('hello', 42)
```

`setTimeout()`

What: Runs a function after a delay in ms

Use: *Delayed actions, debouncing*

```
setTimeout(()=>fn(), 1000)
```

`setInterval()`

What: Runs a function repeatedly at interval ms

Use: *Clocks, polling, animations*

```
setInterval(()=>tick(), 1000)
```

`clearTimeout()`

What: Cancels a pending setTimeout

Use: *Cancel scheduled actions*

```
const id=setTimeout(fn,1000);
clearTimeout(id)
```

`clearInterval()`

What: Cancels a running setInterval

Use: *Stop clock or polling loop*

```
const id=setInterval(fn,1000);
clearInterval(id)
```

`fetch()`

What: Makes HTTP request, returns a Promise

Use: *Call APIs, load data*

```
const res=await fetch('/api');
const data=await res.json()
```

`encodeURIComponent()`

What: Encodes special characters for safe URL use

Use: *Build query strings safely*

```
encodeURIComponent("a b") // "a%20b"
```

`decodeURIComponent()`

What: Decodes an encoded URI component

Use: *Read URL parameters*

```
decodeURIComponent("a%20b") // "a b"
```

`structuredClone()`

What: Creates a deep clone of an object

Use: *Copy objects without references*

```
const b=structuredClone({a:{c:1}})
```

`isNaN()`

What: Returns true if value is Not a Number

Use: *Quick NaN guard (prefer Number.isNaN)*

```
isNaN('hello')          // true
```

Quick Cheat Sheet

Most used one-liners

Task	Code
Double all numbers in array	`[1,2,3].map(x => x * 2)`
Filter even numbers	`[1,2,3,4].filter(x => x % 2 === 0)`
Sum all numbers	`[1,2,3].reduce((a,b) => a+b, 0)`
Find item in array	`arr.find(x => x.id === target)`
Remove duplicates	`[...new Set(array)]`
Sort numbers ascending	`arr.sort((a,b) => a - b)`
Round to 2 decimal places	`(3.14159).toFixed(2)`
Random integer 0-9	`Math.floor(Math.random() * 10)`
Check if value is number	`!isNaN(value) && typeof value === 'number'`
Deep copy an object	`structuredClone(obj)`
Convert object to JSON	`JSON.stringify(obj, null, 2)`
Get all object keys as array	`Object.keys(obj)`
Pad number to 2 digits	`String(n).padStart(2, '0')`
Get current year	`new Date().getFullYear()`
Split sentence into words	`"hello world".split(" ")`